4.4.5 Consider dermal or skin substitutes as an adjunct to standard care when treating diabetic foot ulcers, only when healing has not progressed and on the advice of the multidisciplinary foot care service. NICE clinical guideline NG19 (Adopted)

4.4.6 1f Consider negative pressure wound therapy after surgical debridement for diabetic foot ulcers, on the advice of the multidisciplinary foot care service. NICE clinical guideline NG19 (Adopted)

4.4.7 Do not select agents reported to improve wound healing by altering the biology of the wound, including growth factors, bioengineered skin products and gases, in preference to accepted standards of good quality care. (Strong; Low) IWGDF 2015 (Adopted)

4.4.8 Do not select agents reported to have an impact on wound healing through alteration of the physical environment, including through the use of electricity, magnetism, ultrasound and shockwaves, in preference to accepted standards of good quality care. (Strong; Low) IWGDF 2015 (Adopted)

4.4.9 Do not select systemic treatments reported to improve wound healing, including drugs and herbal therapies, in preference to accepted standards of good quality care. (Strong; Low) IWGDF 2015 (Adopted)

4.4.10 Redistribution of pressure off the wound to the entire weight-bearing surface of the foot ("off-loading"). While particularly important for plantar wounds, this is also necessary to relieve pressure caused by dressings, footwear, or ambulation to any surface of the wound. (Strong; High) IDSA 2012 guidelines (Adopted)

4.4.11 When deciding about wound dressings and offloading when treating diabetic foot ulcers, take into account the clinical assessment of the wound and the person's preference, and use devices and dressings with the lowest acquisition cost appropriate to the clinical circumstances. NICE clinical guideline NG19 (Adopted)

4.5 : FOOTWEAR

4.5.1 To heal a neuropathic plantar forefoot ulcer without ischemia or uncontrolled infection in a patient with diabetes, offload with a non-removable knee-high device with an appropriate foot-device interface. (Strong; High) IWGDF 2015 (Adopted)
Non-removable (cast) walker: Same as removable (cast) boot/walker but then with a layer(s) of fibreglass cast material circumferentially wrapped around it rendering it irremovable (also known as "instant total contact cast")

4.5.2 When a non-removable knee-high device is contraindicated or not tolerated by the patient, consider offloading with a removable knee-high walker with an appropriate foot-device interface to heal a neuropathic plantar forefoot ulcer in a patient with diabetes, but only when the patient can be expected to be adherent to wearing the device. (Weak; Moderate) IWGDF 2015 (Adopted)
Removable (cast) boot/walker: Prefabricated removable knee-high boot with a rocker or roller outsole configuration, padded interior, and an insertable and adjustable insole which may be total contact.

4.5.3 When a knee-high device is contraindicated or cannot be tolerated by the patient, consider offloading with a forefoot offloading shoe, cast shoe, or custom-made temporary shoe to heal a neuropathic plantar forefoot ulcer in a patient with diabetes, but only and when the patient can be expected to be adherent to wearing the shoes. (Weak; Low) IWGDF 2015 (Adopted)

4.5.4 Instruct an at-risk patient with diabetes to wear properly fitting footwear to prevent a first foot ulcer, either plantar or non-plantar, or a recurrent non-plantar ulcer. When a foot deformity or a pre-ulcerative sign is present, consider prescribing therapeutic shoes, custom-made insoles, or toe orthosis*. (Strong; Low) IWGDF 2015 (Adopted)
*Toe orthosis:- An in-shoe orthosis to achieve some alteration in the function of the toe.

4.5.5 To prevent a recurrent plantar foot ulcer in an at-risk patient with diabetes, prescribe therapeutic footwear that has a demonstrated plantar pressure relieving effect during walking and encourage the patient to wear this footwear. IWGDF 2015 (Adapted)

4.5.6 Instruct a patient with diabetes not to use conventional or standard therapeutic footwear to heal a plantar foot ulcer. IWGDF 2015 (Adapted)
Explanatory note: Use footwear with following features -
Sandals: should have adjustable straps, insole, full heel counter and rigid outsole.
Shoes: should have wide toe box extra depth and without laces.

4.5.7 Consider using shoe modifications, temporary footwear, toe spacers or orthoses to offload and heal a non-plantar foot ulcer without ischemia or uncontrolled infection in a patient with diabetes. The specific modality will depend on the type and location of the foot ulcer. (Weak; Low) IWGDF 2015 (Adopted)

4.5.8 If other forms of biomechanical relief are not available, consider using felted foam* in combination with appropriate footwear to offload and heal a neuropathic foot ulcer without ischemia or uncontrolled infection in a patient with diabetes. (Weak; Low) IWGDF 2015 (Adopted)
Felted foam - A fibrous, unwoven material backed by foam with absorbing and cushioning characteristics. The foam is generally 'rubber foam' or 'PU foam' which is formed by either a polyester or polyether polyol resin, in conjunction with water and Toluene di Isocyanate, along with various catalysts and blowing agents and colouring pigments to give the desired compression.

4.6: Diabetic foot with osteomyelitis and CHARCOT'S FOOT

4.6 Treatment of Diabetic foot with osteomyelitis

4.6.1 For an infected open wound, perform a probe-to-bone test; in a patient at low risk for osteomyelitis a negative test largely rules out the diagnosis, while in a high risk patient a positive test is largely diagnostic. (Strong; High)IDSA 2012 guidelines (Adopted)

4.6.2 Markedly elevated ESR is suggestive of osteomyelitis in suspected cases. IWGDF 2015 (Adapted)
Explanatory note: Tests for serum inflammatory markers are costly and not widely available, except ESR. Also these tests are not diagnostic of DFO.

4.6.3If osteomyelitis is suspected in a person with diabetes but is not confirmed by initial X-ray, consider an MRI to confirm the diagnosis. In places where MRI is unavailable, diagnose osteomyelitis by the PTB test (clinically) and/or taking a Bone biopsy and culture. (Adapted from NICE clinical guideline NG19 and Expert group discussion)
Explanatory note: Expert Consensus says that as availability of MRI is limited across the country, it is recommended to use MRI wherever available. At the primary and secondary health centre levels, the PTB test and bone biopsy and culture are more feasible and economical and reasonably accurate.